



GASES



Gases are all around us and can't usually be seen or felt, but some can be smelled. All smells are molecules of gas mixed in the air. If you smell some tasty soup, you are actually sniffing in molecules of the soup. When you heat a liquid, it turns into a gas. The heat makes the molecules in the liquid move around faster and faster. The gas molecules fly off in all directions, spreading through the air.



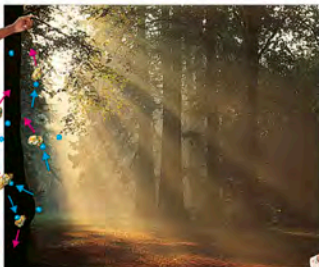
Lighter than air

When a gas gets hot, it takes up more room and gets lighter. A hot-air balloon rises up in the air because the air inside it is hot, and much lighter than the air outside.



Molecules in a gas

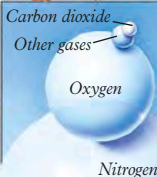
Each child is acting like a molecule in a gas. Look how they are bumping into each other.



Dancing on air

When a shaft of light shines through a gap in the trees, specks of dust look like they are dancing in the sunlight. What is really happening is that the molecules in the air currents move around, flicking the dust in all directions.

There are no links between any of the molecules in a gas.



In the air

There are a number of different gases in the air we breathe. 78 percent of air is nitrogen gas. Oxygen makes up 21 percent. Carbon dioxide and a variety of other gases make up just one percent of the gases in the air.

As a gas spreads, the molecules get farther and farther apart. Gases will always spread to fit the space they are in.

